Incidence invariants for the Higman–Sims geometries

L091 classification component

Let $G = \mathrm{HS}$ in the acquired permutation representation of degree 100. Its order is 44 352 000. We use the complete subgroup and panel inventory and the separately reviewed enumeration of the firm, residually connected, flag-transitive, residually weakly primitive, locally two-transitive IP2 coset geometries for this action. In ranks at least two, that enumeration gives 130 families and 86 classes under $\mathrm{Aut}(G)$ and permutations of the types. The rank counts are respectively $(6, 44, 67, 12, 1)$ and $(6, 30, 41, 8, 1)$ for ranks two through six. There are no higher ranks. The acting automorphism group is represented faithfully by the acquired normalizer of order 88 704 000.

The enumeration and its completeness are inputs here. In particular, the present argument must still exclude an incidence isomorphism which does not preserve the specified action of G. We refer to the numbered representatives in the fixed, acquired file `full_fusion03/result.g`. These numbers are internal identifiers, not the source's diagram labels.

Lemma 1. *Suppose a group G acts transitively on each type of a finite typed incidence structure. Let X be a type and let $S_y \subseteq X$ be the set of objects incident with an object y of another type. If $G_{S_y} = G_y$, the map $gy \mapsto gS_y$ is injective. More generally, suppose the objects of each remaining type can successively be identified by their sets of neighbours in types already identified. Then restriction to X is faithful on the group of automorphisms preserving these types.*

Proof. The fibres of the equivariant map $G/G_y \longrightarrow G/G_{S_y}$ have size $[G_{S_y} : G_y]$. Equality of the stabilizers proves the first claim. An automorphism fixing X pointwise fixes every distinct subset of X, hence fixes the next type pointwise. Induction proves the second assertion. This argument does not require that G be the full automorphism group. □

Lemma 2. *Let Γ and Γ' have a distinguished type X and X', recognizable from their type sizes. Suppose each structure defines a graph on that type purely by incidence, restriction of its automorphism group to that type is faithful, and the image of its specified group G is the derived subgroup of the full graph automorphism group. Every incidence isomorphism $\Gamma \longrightarrow \Gamma'$ conjugates the specified G actions.*

Proof. An incidence isomorphism f induces a graph isomorphism f_X. It conjugates the full graph automorphism groups and consequently their derived subgroups. For $g \in G$, choose $h \in G$ whose action on X' equals that of fgf^{-1}. The automorphism $h^{-1}fgf^{-1}$ is trivial on X', so faithfulness makes it trivial on all of Γ'. This gives the asserted conjugation of the entire actions. □

Theorem 1. *Under the acquired enumeration and the exact computational inputs specified below, the* 86 *representatives are pairwise nonisomorphic as incidence geometries, even allowing permutations of types. There is one additional isomorphism class in rank one, consisting of a set of* 176 *objects. It is a truncation of a higher rank geometry. Thus the numbers of isomorphism classes in ranks one through six are*

$$1, \quad 6, \quad 30, \quad 41, \quad 8, \quad 1.$$

Proof. Write G_i for the stabilizers of a base chamber. Since the action is flag-transitive, the number of flags whose set of types is a nonempty subset J is $|G|/|\bigcap_{j\in J} G_j|$. These counts are incidence invariants, with their coordinates permuted when the types are permuted. In particular the multiset of orders $|G_i|$, the multiset of orders $|G_i \cap G_j|$, and the Borel order are invariants for these fixed-order actions. Computing these from the actual subgroups separates all representatives except the pairs $(14, 15)$ and $(83, 84)$. For the rank five pair $(83, 84)$, the vector of all nonempty flag stabilizer orders, minimized over every permutation of the five types, has different values. These are checked by intersections of the acquired subgroups, not inferred from the printed diagrams.

It remains to separate 14 and 15. Their three stabilizer orders are $192, 1920, 40320$. Their type sizes are therefore $231000, 23100, 1100$, so every isomorphism preserves each type. Call these types lines, blocks and points, respectively, and let P, Q, R denote their stabilizers at a base chamber. Take the actual action on the 1100 right cosets of R. It is faithful. If $z = R$, the shadows of the base line and block on the point type are z^P and z^Q, of sizes two and sixteen.

Define two points to be adjacent when some line is incident with both. This graph is intrinsic to the geometry. To construct it, let w be the other member of z^P. Its neighbours at z are exactly w^R: the lines incident with z form the R orbit of the base line. Transporting this neighbour set by a representative of every coset gives the whole graph. In both cases it has 1100 vertices and valency 105.

The stabilizer in G of the base block shadow z^Q has order 1920, equal to $|Q|$. Thus blocks are determined by their sixteen point shadows. Lines are *not* determined by their two point shadows: the corresponding fibre has size four. Instead consider the set $\mathcal{B} = (z^Q)^P$ of block shadows neighbouring the base line. There are four. The group P is transitive on them, and the intersection index $[P : P \cap Q] = 4$ confirms that they are all its incident blocks. Exactly 48 of the 1920 elements of Q stabilize $\mathcal{B}$ setwise. Put $T = G_{\mathcal{B}}$. We have $P \leq T$, and the orbit of z^Q under T is all four members of $\mathcal{B}$, since P is already transitive. Also $T_{z^Q} = T \cap Q$, by the block stabilizer equality. Hence

$$|T| = 4 \cdot 48 = 192 = |P|.$$

The four neighbouring blocks therefore determine each line. Lemma 1 proves faithfulness of restriction of the full geometry automorphism group to the point type.

The complete graph automorphism computation returns order 88 704 000 for each of these two graphs. Every returned generator is checked against every adjacency. In GAP the group D generated by these permutations has this exact order, contains the actual coset image A of G, and satisfies $D' = A$. Completeness of the graph computation is the explicitly retained nauty input; the subgroup equality $D' = A$ is tested on the actual permutations. Lemma 2 now shows that any incidence isomorphism between these geometries would conjugate their full specified G actions. Such an isomorphism induces an automorphism of the abstract group G. After composing with an element of G to move the image chamber back to the base chamber, it sends the unordered family of parabolic stabilizers onto the other family. The acquired complete $\mathrm{Aut}(G)$ comparison excludes this for representatives 14 and 15. This proves that all 86 representatives are distinct.

Finally, the complete panel inventory has exactly two two-transitive actions in rank one, at subgroup anchors 586 and 587. Both completed enumeration records have levels $(1, 0)$, and both actions have degree 176, are faithful and are two-transitive. A rank one geometry has no incidence between distinct types, so its isomorphism type depends only on its number of objects. It has full automorphism group Sym(176). The retained conjugators in the acquired normalizer identify each point stabilizer with a parabolic of a higher rank representative. They give explicit truncation isomorphisms. There is consequently one rank one class, and none which is not a truncation. Together with the acquired exhaustion in higher ranks, this proves the theorem. □

Attribution and limits. The counts recover the classification asserted by Dimitri Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33

(2005), 231–236, in the fixed local six-page edition. The complete subgroup catalogue, the panel enumeration and the full automorphism action retain the attributions and computational premises of their acquired reviews. The shadow lemmas are elementary consequences of equivariance, orbit–stabilizer and characteristic subgroups. Their application above supplies the missing distinction between equivalence of group actions and isomorphism of the geometries; it does not assert a new classification or a solution to a new open question.

The exact retained computational inputs are GAP's permutation group, coset action and subgroup algorithms and nauty's complete graph automorphism algorithm. The supplied generators, adjacency tests, stabilizer and derived subgroup equalities are additional finite checks, not a substitute for nauty completeness. No Lean formalization is claimed. The higher rank truncation counts, the supplementary diagrams and full automorphism data, and the source's two Hoffman–Singleton constructions remain separate obligations. This component therefore gives no whole-manuscript completion or compression measurement.